AN EFFICIENT INTRUSION DETECTION SYSTEM FOR INTERNET OF THINGS (IOT) ENVIRONMENTS USING FEATURE SELECTION AND ENSEMBLE SOFT VOTING CLASSIFIER

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AUGUST, 2024

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FUKU/SCI/19/COM/0231

PROJECT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE, FACULTY OF SCIENCE, FEDERAL UNIVERSITY OF KASHERE, IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF SCIENCE (B.Sc.) IN COMPUTER SCIENCE

AUGUST, 2024

DECLARATION

I declare that the work in this project dissertation titled **an Efficient Intrusion Detection System for Internet of Things (IoT) Environments Using Feature Selection and Ensemble Soft Voting Classifier** has been performed by me in the Department of Computer Science, under the supervision of Mr. Jeremiah Isuwa. The information derived from the literature has been duly acknowledged in the text and a list of references is provided. No part of this project dissertation was previously presented for another degree or diploma at this or any other Institution.

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GOMBE STATE, NIGERIA

Signature and Date

CERTIFICATION

This project dissertation titled **AN EFFICIENT INTRUSION DETECTION SYSTEM FOR INTERNET OF THINGS (IOT) ENVIRONMENTS USING FEATURE SELECTION AND ENSEMBLE SOFT VOTING CLASSIFIER** by **Godswill Enefiok Archibong (FUKU/SCI/19/COM/0231)** meets the regulations of the degree of Bachelor of Science (B.Sc.) in Computer Science of the Federal University of Kashere, Gombe State and is approved for its contribution to knowledge and literary presentation.

Mr. Jeremiah Isuwa _____

Supervisor Signature and Date

Mr. M. B. Jibrin _____

Head of Department Signature and Date

…… _____

External Examiner Signature and Date

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DEDICATION

This work is dedicated to God Almighty, whose grace and guidance have been my constant source of strength and inspiration. I also dedicate this work to my parents, with special recognition to my beloved mother, Mrs. Gift Ufford whose unwavering love, support, and sacrifices have been the foundation of all my achievements.

ABSTRACT

The exponential growth of the Internet of Things (IoT) has transformed various sectors by enhancing efficiency, data collection, and automation. However, this proliferation has also introduced significant security challenges, as IoT devices are often equipped with minimal security features, making them prime targets for cyber-attacks. Traditional Intrusion Detection Systems (IDS) have proven inadequate for IoT environments due to their inability to manage the vast and diverse data generated by these devices and to detect complex, evolving threats. Thus this research addresses these challenges by developing an efficient, machine learning-based IDS specifically tailored for IoT networks. Utilizing the NSL-KDD dataset, a benchmark for intrusion detection, this dissertation focuses on employing feature selection using a combination of Chi-square and Mutual Information filter methods to increase the effectiveness and accuracy of the detection process by selecting the most relevant features from the dataset. Utilizing the complementing strengths of the Random Forest (RF), Logistic Regression (LR), and Artificial Neural Network (ANN) algorithms, the proposed method leverages an ensemble soft voting classifier to develop and improve the overall performance of the IDS. The experimental results demonstrated that the proposed method while trained with the feature-selected dataset performed better achieving an accuracy of 99.40% and an F1 score of 99.44%, compared to the slightly lower performance when trained on the entire dataset highlighting the significant impact of feature selection on enhancing the IDS's performance. The superior performance of the proposed method is further demonstrated by comparison with existing studies in the literature. These findings indicate that feature selection with an ensemble soft voting technique is critical in developing high-performance detection systems for IoT networks.

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NOTATIONS

ANN:	Artificial Neural Network
DR:	Dimensionality Reduction
FE:	Feature Extraction
FS:	Feature Selection
IDS:	Intrusion Detection System
IoT:	Internet of Things
LR:	Logistic Regression
ML:	Machine Learning
RF:	Random Forest

CHAPTER ONE

INTRODUCTION

1.1. Background to the Study

The advancement in computer technologies such as mobile and pervasive computing, internet applications, and services has led to the proliferation of Internet of Things (IoT) devices (Zeddini et al., 2022). IoT is a network of devices that allows these devices to share information directed towards different purposes (Stewart et al., 2020). Such devices include desktops, laptops, smartphones, and tablets. The IoT is an excellent platform to connect smart items globally over the Internet without requiring human interference, yet like any network, it is vulnerable to cyberattacks (Aboubakar et al., 2022). With the increasing demand and growth in the IoT automated network system, the IoT models are getting complicated day by day (Aboubakar et al., 2022). Users can remotely manage machines from anywhere, making them vulnerable to lots of threats. So, the security of the IoT system is very concerned about the increasing number of smart devices today because the devices contain private and important user information (Williams et al., 2022).

In recent years, there has been a significant increase in the usage of IoT devices and networks across various sectors such as healthcare, urban development, logistics, and agriculture (Schiller et al., 2022). With this extended use of IoT, new protocols are being deployed (Hindy et al., 2020). Thus, IoT technology guarantees to improve and help our personal, professional, and societal lives (Abdel-Basset et al., 2022). According to Statista, (2016) reports, the number of IoT devices has been increasing annually, and by 2025, it is expected to reach 75.44 billion. Such a massive surge of IoT devices ultimately results in more attackers targeting IoT networks.

Reports state that most of the attack traffic generated on IoT networks is automated through various means such as scripts and malware (Sarwar et al., 2023). Intrusions occur in a matter of seconds in today's world. Consequently, a more robust security system becomes necessary. An Intrusion Detection System (IDS) is an effective technique for detecting cyber-attacks in any network. Most of the latest IDS are based on a Machine Learning (ML) algorithm for the detection of cyber-attacks in the network. currently, their use is in all areas of human life (Siyyal et al., 2022).

IDSs in IoT are designed to detect and protect against unauthorized access or malicious activities on a network, system, or application. In literature, different techniques from various disciplines have been employed to develop efficient IDS but such techniques generally have some shortcomings (A. Ashok Kumar, 2023; Asmat, 2023; Villa-Henriksen et al., 2020). Traditional intrusion prevention techniques, such as firewalls, access control or encryption, and password-based security have failed to fully protect networks and systems from increasing, sophisticated attacks and malware (Zeddini et al., 2022). They can be used to detect and respond to any malicious activity related to the IoT. They can also help identify attacks and potential breaches, allowing an organization to take appropriate action before damage is done.

The current trend in the research community to detect intrusion is based on ML (Almomani et al., 2021; S. Das et al., 2022; Saranya et al., 2020). This technique has the capability of autonomous packet detections with high detection rates and low false positive rates while the system quickly adapts itself in a dynamic environment (Tahsien et al., 2020). ML algorithms can analyze large amounts of data to identify patterns that could indicate attacks. These algorithms identify patterns in the data and search for anomalies or malicious activities using a variety of methods, including supervised learning, unsupervised learning, and reinforcement learning. This

allows systems to be better equipped to understand and defend against malicious activity. While the integration of ML techniques has significantly advanced IDS, the effectiveness of these models heavily relies on the quality of the datasets used for training. The optimum feature set needs to be identified by extracting the unnecessary features which results in a reduction in processing time and gives higher detection accuracy in IDS (A. Ashok Kumar, 2023). ML can be applied to IDSs to help improve the system's ability to run autonomously and increase the accuracy of the system when raising the alarm on a suspected attack (Alsahli et al., 2021).

This research aims to enhance IoT security by creating an efficient IDS using advanced ML techniques. The integration of a soft voting classifier model with feature selection is expected to provide an efficient and reliable solution that addresses important security issues in IoT networks.

1.2. Problem Statement

IoT continues to expand, integrating into various aspects of daily life and industrial processes, the security of these interconnected devices has become a pressing concern. Despite the numerous advantages IoT offers, such as improved efficiency and enhanced data collection, it also exposes networks to significant vulnerabilities. Many IoT devices are designed with minimal security features, making them susceptible to cyber-attacks.

Traditional IDS are often inadequate for IoT environments due to their inability to handle the vast and diverse data generated by these devices. These systems struggle with detecting complex, evolving threats and adapting to the dynamic nature of IoT networks. In 2016, a series of distributed denial of service attacks, targeting loT networks and websites like Twitter, Netflix,

and PayPal pointed out to the world how urgent it is to develop a strong security solution. (Shurman et al., 2020). This inadequacy leaves IoT ecosystems vulnerable to intrusions that can compromise data integrity, privacy, and overall operational stability.

The problem, therefore, is the lack of an effective and adaptive intrusion detection mechanism capable of securing IoT networks against sophisticated cyber threats. There is an urgent need for advanced security solutions that can learn and evolve with the changing threat landscape.

Thus, this research seeks to address this critical issue by developing a machine learning-based intrusion detection system tailored specifically for IoT environments. Utilizing the NSL-KDD dataset, a widely recognized benchmark for intrusion detection research, the proposed solution will focus on optimizing feature selection to enhance the efficiency and accuracy of the detection process. By identifying the most relevant attributes from the dataset, the system aims to improve model performance and reduce computational overhead, providing a robust defense against potential intrusions and ensuring the safety and reliability of IoT networks.

1.3. Significance of the Study

While IoT technologies play a vital part in improving real-life smart systems, like smart cities, smart homes, and smart healthcare, the large-scale and ubiquitous nature of IoT systems has introduced new security challenges (Sarwar et al., 2023). On top of these security challenges, IoT devices cannot afford the implementation of advanced security features because of their restricted energy and computation resources. Due to the interconnected and interdependent settings of the IoT, new attack surfaces are emerging very regularly (Al-Garadi et al., 2020). Thus, IoT systems are more vulnerable as compared to traditional computing systems. This

necessitates research in specific detective and preventive techniques for IoT systems to protect against IoT device-based threats. For protecting IoT systems against cyber threats, another line of defense should be developed in IoT networks. IDSs fulfill this purpose. This research work covers research work on IDSs for IoT systems (Imad et al., 2022)

1.4. Aim and Objectives

This project aims to develop a soft voting classifier-based intrusion detection system tailored specifically for Internet of Things environments.

The specific objectives are as follows;

i. Identify significant features from the NSL-KDD dataset using Mutual Information and Chi-square filters.
ii. Construct a predictive model using a soft voting classifier with three algorithms; Artificial Neural Network (ANN), Random Forest (RF) and Logistic Regression (LR).
iii. Evaluate the performance of the constructed model with existing methods based on key metrics such as F1 Score, recall, precision, and accuracy, using the ID dataset from Kaggle.

1.5. Dissertation Organization

The rest of the dissertation will be organized as follows:

Chapter Two: It explores previous research and methodologies used in developing IDS, focusing on their application within IoT environments. Additionally, the chapter provides a detailed explanation of key concepts and technologies relevant to the dissertation, including ML algorithms, feature selection techniques, and IoT. Finally, reviews existing literature related to Machine Learning for IDS in IoT.

Chapter Three: describes the proposed methodology used in the research work. It outlines the steps involved in dataset acquisition, preprocessing, and feature selection. Additionally, it provides a detailed explanation of the ANN model design and implementation, including parameter optimization techniques. The chapter also discusses the experimental setup and data collection process for evaluating the performance of the IDS.

Chapter Four: presents the results obtained from the experiments conducted based on the proposed methodology. It provides a comprehensive analysis of the performance of the constructed IDS, comparing it with existing methods using key metrics such as F1 Score, recall, precision, and accuracy. The chapter also discusses the findings of the research objectives and identifies any limitations encountered during the experimentation.

Chapter Five: concludes the dissertation by summarizing the key findings and insights obtained from the research. Furthermore, this chapter outlines the scope for future research in the field, providing recommendations for further exploration and improvement of Machine Learning-based IDS in IoT environments.

CHAPTER TWO

LITERATURE REVIEW

2.1. Introduction

This chapter serves to introduce key concepts necessary for understanding both the existing and proposed methodology. It commences by giving a brief overview of IoT systems and the possible types of attracts that threaten the system. It further describes the notion of feature selection, encompassing various types such as filter, wrapper, and embedded methods. Following this, the chapter explains the concept of ML, offering examples of its diverse algorithms. Finally, this chapter gives a comprehensive review of the existing literature related to Machine Learning techniques for IDS in IoT environments.

2.2. Dimensionality Reduction (DR)

Datasets' growth and update speed are accelerating, and the data is developing in a high-dimensional and unstructured direction. Massive and complex data contains a lot of useful information, but it also increases the difficulty of using the data effectively. Datasets with many features are called high-dimensional data (Capitaine et al., 2021). Theoretically, having a large amount of data with more attributes and features should result in more discriminating power and more accuracy. But practically, many machine learning algorithms have shown that this is not always true (Das et al., 2022). For example, the problem called the "curse of dimensionality" appears due to the rapid and large-scale expansion of dimensions (Hutzenthaler et al., 2022). A substantial amount of computing time and storage space is dedicated to processing data, with valuable information often buried within its complexity, making it difficult to discover the

essential features of the data. It takes lots of time and manpower to process the data. This problem also has a bad influence on the accuracy of prediction. In Figure 2.1, taking the performance of the classifier as an example, when the data dimension increases, the performance of the classifier becomes better; when the data dimension continues to increase, the performance of the classifier becomes worse. To address this issue, DR techniques are employed. These methods aim to reduce noise and enhance the efficiency of learning algorithms by transforming data into a lower-dimensional space. By doing so, they alleviate the effects of the dimensionality curse (Isuwa et al., 2022a).

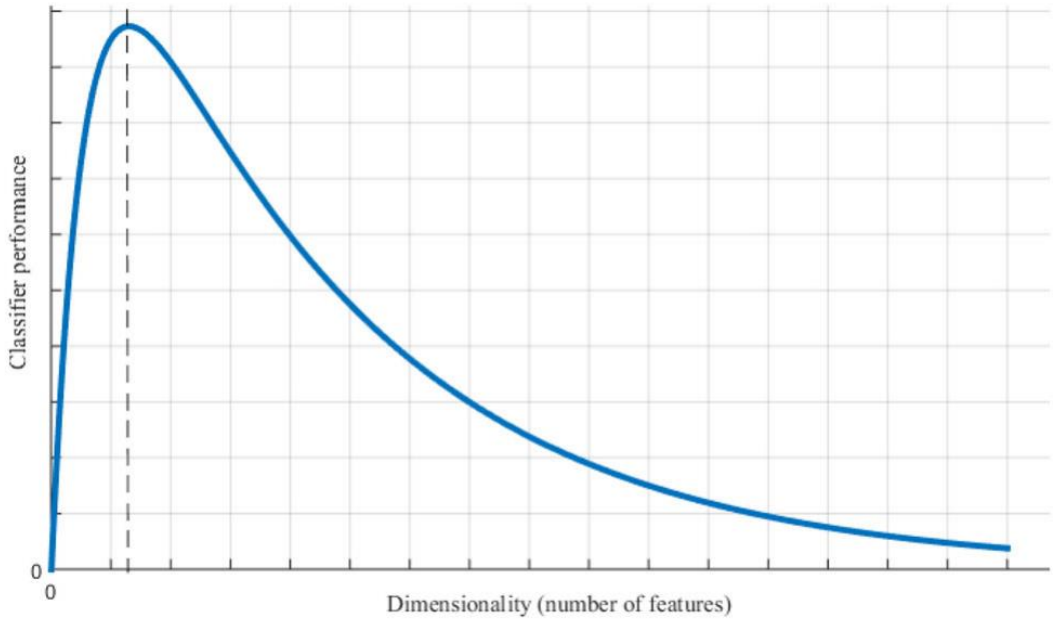

Figure 2.1: The trend of classifier performance with the change of dimension (Jia et al., 2022)

DR can be seen from two main perspectives: feature extraction and feature selection.

i. Feature Selection: Feature selection involves selecting a subset of the original features that are most relevant to the problem at hand for final classification. The goal is to reduce the dimensionality of the dataset while retaining the most important features.

ii. Feature Extraction: Feature extraction involves creating new features by combining or transforming the original features. The goal is to create a set of features that captures the essence of the original data in a lower-dimensional space. Feature extraction is a technique used to transform existing features from input data, often utilizing linear methods like Principal Component Analysis (PCA) and non-linear methods like Kernel PCA (Isuwa et al., 2022a).

The two DR techniques can be approached either separately or in combination. In specific scenarios, they are treated as distinct methods Figure 2.2 gives the hierarchical structure of DR showing its different methods.

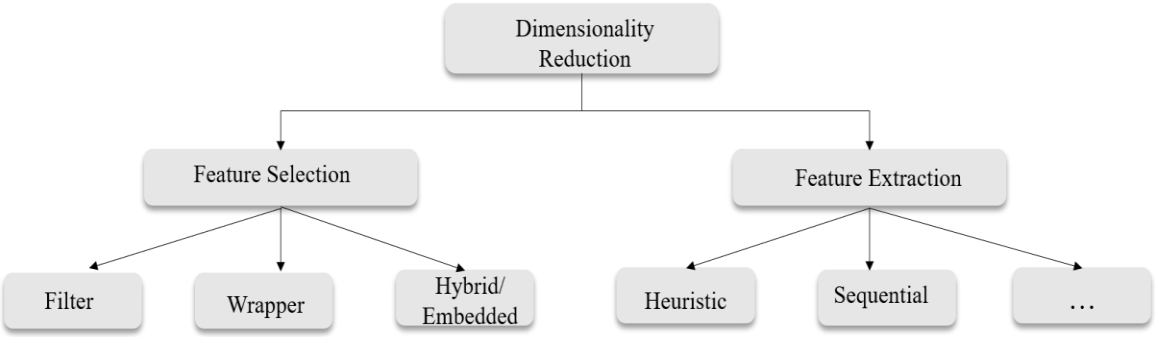

Figure 2.2: Methods of Dimensionality Reduction (Isuwa et al., 2022a).

2.2.1. Benefits of dimensionality reduction

Beyond just saving processing time and storage space, DR has many other benefits (Palo et al., 2021). Notably: DR is used to prevent overfitting and improve data visualization. It simplifies high-dimensional data, thus avoiding potential issues with poor generalization performance, and assists in data preprocessing before passing it to algorithms, which enhances efficiency and accuracy in data analysis. This results in improved performance by reducing noise and irrelevant

information in the data. Overall, dimensionality reduction is a crucial step in machine learning model development.

2.3. Feature Selection (FS)

FS can also be called variable selection or feature subset selection, and it is a process of selecting feature subsets of relevant features that are applied to model construction and fully describe the given problem with minimum degradation of performance (Lyu et al., 2023; Nanda & Patra, 2020). The most important prerequisite for using feature selection techniques is that the dataset contains many redundant or related features that can be deleted without losing significant information (Sharma & Mishra, 2022).

In the feature selection process, the algorithm automatically searches for the best subset of attributes in the dataset. A subset of the dataset is provided to the algorithm for subset generation. The loop is implemented until it selects a sufficient number of features from the dataset without influencing system performance and efficiency. A subset evolution function is used to track this activity of the algorithm. The whole process of feature selection is shown in Figure 2.3.

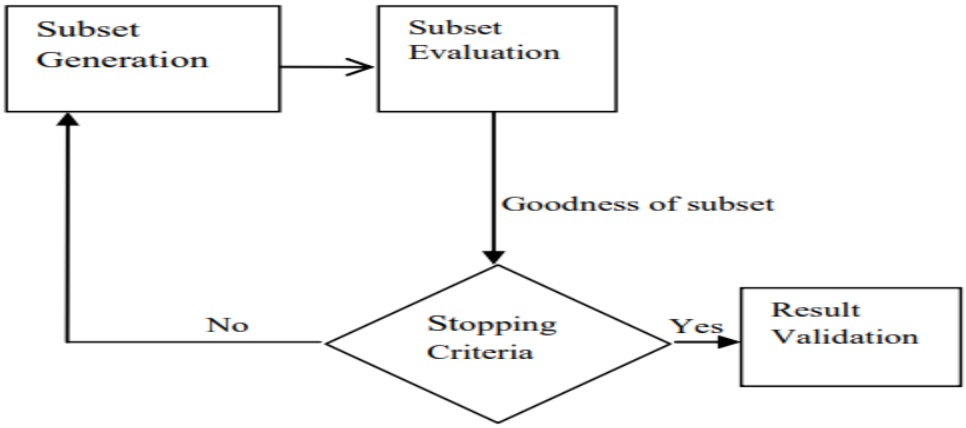

Figure 2.3: Procedure for Feature Selection (Jia et al., 2022)

The FS algorithm uses computing principles to select the features. This algorithm is divided into two phases; Subset Generation and Subset Evaluation. In subset Generation, the desired subset is generated from the input dataset. Subset Evaluations are used to check whether the generated subset is optimal or not. These operations can be done in three ways; Filtering, wrapper, and Embedded.

2.3.1. Filter method

The Filter method considers evaluation criteria as the standards for selecting features. It uses feature rank as the standard criteria for feature collection by arrangement (Jia et al., 2022). Fan et al., (2023) mentioned that the filter technique selects a feature based on its overarching attributes, including correlation, class separation, and statistical interdependence. This method usually finds a suitable standard to evaluate variables and uses a threshold to remove variables below this threshold to filter out less relevant variables, reduce the degree of correlation between features, and increase the degree of correlation between features and classes. The filter method is simple and practical, and it is widely used (Solorio-Fernández et al., 2022), However, it does not take into account feature interaction (Iqbal et al., 2020a). Consequently, it may neglect important features that may seem insignificant individually but become significant when combined with other features (Bommert et al., 2020). The filter method process is shown in Figure 2.4.

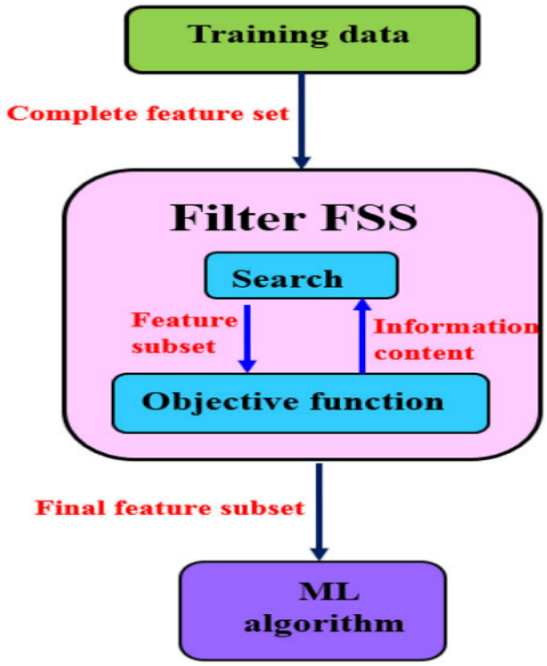

Figure 2.4: The Filter Method Process (Jia et al., 2022)

The Filter method utilizes statistical techniques like correlation analysis (Deng et al., 2022), relief (Jun Dou et al., 2022), mutual information (Song et al., 2021), and chi-square (Isuwa et al., 2023) to assess and rank individual features according to their importance. Each feature is assigned a score indicating its level of importance, with higher scores indicating greater significance (Isuwa et al., 2023).

2.3.1.1. Chi-Square

The Chi-square test, as described by Bahassine et al., (2020) is a statistical technique employed to evaluate the independence of two categorical variables within a dataset. This technique examines the distinction between the observed and predicted frequencies to determine whether or not there is a significant relationship between the variables. The Chi-square test is widely used

in the field of FS to identify the elements that are most important for a certain classification problem. The formula for the Chi-square test statistic is shown in Equation 2.1:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \tag{2.1}$$

In this context, (O_i) denotes the observed frequency for each category, which is the actual count of occurrences for each category in the dataset. On the other hand, (E_i) signifies the expected frequency, which is computed based on the marginal totals of the variables, under the presumption of independence.

The Chi-square test is used in FS to compute the Chi-square statistic for each feature related to the target variable. Higher Chi-square values indicate greater relevance to the target variable, indicating a stronger association (Isuwa et al., 2023). These features are chosen for further analysis or model construction, especially in datasets with categorical features and target variables.

2.3.1.2. Mutual Information

Mutual information is another statistical method used in FS. It is the measure of how two variables (a, b) are mutually dependent (Zhou et al., 2024). Mutual information is frequently used in FS within the field of ML, where it is employed to measure the degree of dependence between features and the target variable (Maria et al., 2021). Specifically, it gauges the decrease in uncertainty about a particular variable when knowledge about another variable is provided (Li et al., 2020). The formula for mutual information between two variables X and Y is shown in Equation 2.2:

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right) \tag{2.2}$$

In this context, $p(x,y)$ denotes the joint probability function of the variables X and Y, and $p(x)$ and $p(y)$ are the marginal probability distribution functions X and Y respectively.

Within the scope of FS, the concept of mutual information is utilized to quantify the degree of information that a given feature, denoted as X, contributes towards the understanding of the target variable Y. Features that exhibit high values of mutual information are deemed to be more informative and are consequently chosen for subsequent analysis or the construction of models (Zhou et al., 2024). This approach proves to be particularly efficacious when applied to classification tasks involving both categorical and continuous features.

2.3.2. Wrapper method.

The wrapper method incorporates the process of feature selection into algorithm learning (Jia et al., 2022). Unlike the filter method, the wrapper method is based on three component methods: search strategy, predictor, and evaluation function. The subset of features that are evaluated is determined by the search strategy. The predictor can be any classification method and is viewed as a black box. Its performance is used as an objective function to evaluate the feature subset determined through searching algorithms to find the optimal subset. The wrapper is better than the Filter, but it takes more time and requires more computing resources. The wrapper method procedure is shown in Figure 2.5.

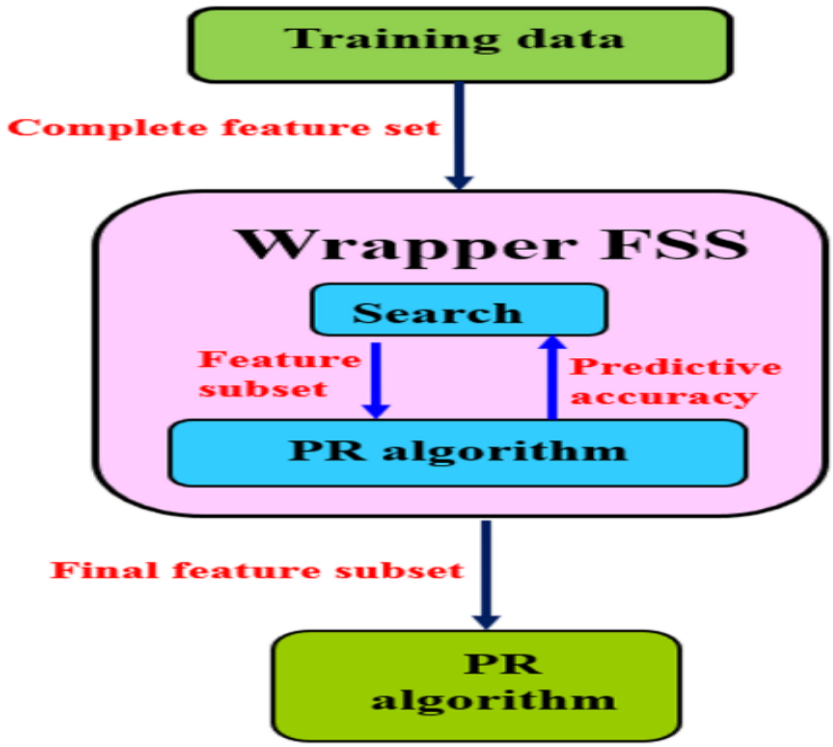

Figure 2.5: Wrapper Method Procedure (Jia et al., 2022)

Wrapper algorithms can be categorized as heuristic or metaheuristic. Heuristic methods involve intelligent guesses during subset selection, which may not always yield optimal results (Isuwa, Abdullahi, & Abdulrahim, 2022). Examples include Sequential Forward and Backward searches. On the other hand, metaheuristics aims to improve the performance of heuristic methods (Isuwa, Abdullahi, & Abdulrahim, 2022). Inspired by natural phenomena, they are considered intelligent algorithms.

2.3.3. Embedded method

The embedded method of feature importance measure is a technique used in ML to evaluate the relevance of features directly within the model training process (Algehyne et al., 2022). It

combines the advantageous aspects of both Filter and Wrapper methods, concurrently managing feature selection and classification (Njeri, 2022).

This method works in a way that the best features are selected during the learning process. The blending of feature selection during the learning process has the advantages of improving computational cost, and classification accuracy and also avoids training the model each time when a new feature is added (Sasikala et al., 2023). Unlike filter methods, which assess features independently of the learning algorithm, embedded methods integrate feature selection into the model's training procedure as shown in Figure 2.6. This approach enables the model to determine the most important features while simultaneously optimizing its performance on the given task (Algehyne et al., 2022).

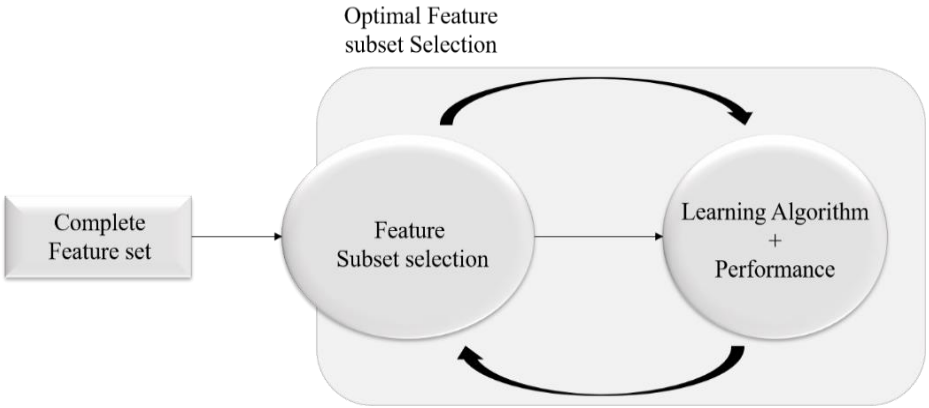

Figure 2.6: The Process of Embedded Method of Feature Selection (Iqbal et al., 2020b)

Common classifiers utilized in embedded FS methods encompass the Random Forest (RF), LASSO, and RIDGE regression, among others (Poojitha & Mahaveerakannan, 2023; Prasanna & Veeramani, 2023).

2.3.4. Hybrid method

The embedded approach and the hybrid method exhibit certain similarities (Isuwa et al., 2022b). The filter approach was initially utilized to minimize the dimension of the feature space, which may generate multiple candidate subsets (Njeri, 2022). Subsequently, a wrapper method is employed to select the best features from these subsets. This approach usually offers better accuracy and computing efficiency, which are characteristics of the wrapper and filter approaches, respectively Isuwa et al., (2022), as seen in Figure 2.8

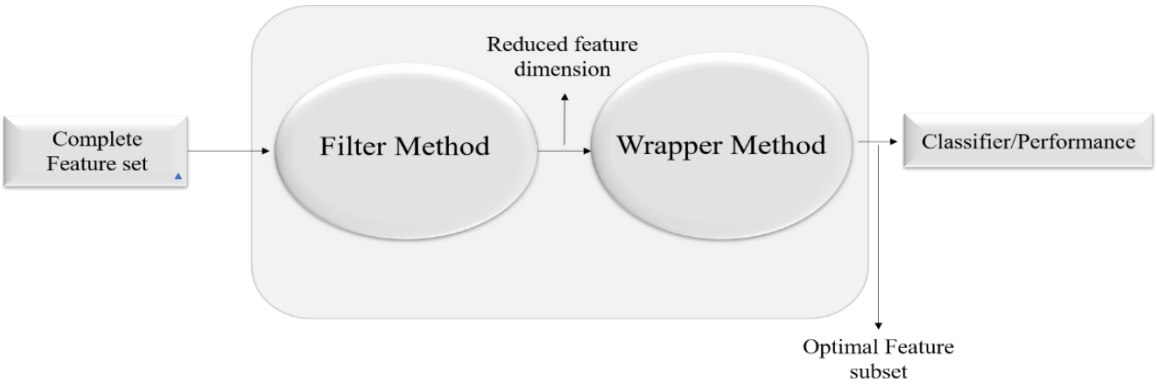

Figure 2.7: The Process of a Hybrid Method of Feature Selection (Isuwa et al., 2022b)

2.4. Feature Extraction (FE)

To avoid the problem of overfitting in any ML model, it is necessary to apply FE techniques. FE aims to reduce the number of features in a dataset by creating new features from the existing ones (Osama et al., 2023). These new reduced set of features should then be able to summarize most of the information contained in the original set of features. In this way, a summarized version of the original features can be created from a combination of the original set. FE implements two functions:

i. Separate effective information from redundant data.

ii. Reduce the operations performed by the classifier by reducing the dimension.

FS has the advantage that the compression of new features is more efficient. However, a notable drawback is that when the original feature set has an obvious physical meaning, the new features may lose meaning (Osama et al., 2023).

2.5. Overview of IoT Systems

In 1982, the Coca-Cola Company introduced the first connected Internet-connected device, a vending machine, to monitor beverage temperature and inventory. Kevin Ashton of Proctor and Gamble coined the term "IoT" in 1999 (A. Ashok Kumar, 2023), but its actual existence was confirmed in 2008. The Coca-Cola Company's vending machine marked the beginning of smart devices in society.

IoT refers to the trillions of physical objects connected to the internet and worldwide storage and data exchange (Yaacoub et al., 2023). The Classification of IoT (Galeano-Brajones et al., 2020) proposed an IoT approach for the integration of various sensors and objects that can communicate with each other without the necessity of human intervention. IoT is described as inclusive of sensory devices that monitor and gather all types of information on both machine and human social life. According to (Thomas et al., 2023), digital information has become a social infrastructure in our society. The perception of IoT is because of the interconnection of devices to one similar network, which is the Internet.

IoT is divided into three layers: the application layer, the perception layer, and the network layer (Paul, 2022). The application layer in IoT is the uppermost layer, which is visible to the end-users of IoT devices. Because this layer interacts with end-users directly, data theft and privacy are key problems (Tewari & Gupta, 2020). The perception layer consists of sensors and actuators (Aarika et al., 2020). This layer is tasked with the collection of information. Sensors sense their surroundings, whereas actuators operate as controllers, taking action depending on the data they collect. It includes the perception nodes as well as the perception network for collecting, processing, and transmitting data from sensors to the higher layers for analysis and decision-making. Finally, the network layer handles data transmission and creates an accessible environment for the perception layer. It employs various protocols like Wi-Fi, Bluetooth, and cellular networks to ensure seamless connectivity between perception nodes and other IoT components (Imad et al., 2022). This layer optimizes data exchange, supports scalability, and ensures data integrity and privacy.

Connectivity to the Internet is the biggest attraction for IoT. The integration of IoT has led to the establishment of the web economy. This is mainly because the Internet broke the physical barriers that limited trade to a regional basis only.

2.5.1. IoT security challenges

In the 21st century, IoT device security has become a pressing issue. On one hand, IoT connects the entire world, bringing it closer together. On the other, it opens various windows for attacks of several types (Tahsien et al., 2020; Zeebaree et al., 2020). The complexity of IoT has resulted in the magnification of the security challenges that had initially been associated with the Internet. Node access, which is the basic functionality of the IoT, is a predisposing factor to the challenges

that face this entity. Security features vary between IoT and conventional networks, with a notable distinction being IoT's reliance on low-power and Lossy Networks (LLNs) (A. Ashok Kumar, 2023). This reliance exposes IoT to potential data loss from node impersonation. IoT also faces security challenges like masquerading and man-in-the-middle attacks (Sadhu et al., 2022). These threats can intercept information within the network and transmit counterfeit data to nodes. Compounding the issue, IoT lacks unified standards that could serve as a foundation for addressing security challenges effectively (Pavani & Rao, 2020). Thus, an advanced kind of security system must be deployed to secure IoT products. (Sadhu et al., 2022) grouped the threats that are related to IoT based on the IoT Layer, which are perception, network, and application which is summarized in Table 2.1.

Table 2.1: Various cyber-attacks against IoT devices (Islam & Aktheruzzaman, 2020)

Classification	**Security Attacks**	**Security Impacts**
Physical Attacks	Node tempering, Side-channel Analysis, Radiofrequency jamming	Hackers can physically damage and corrupt an IoT device using various techniques.
Network Attacks	Traffic analysis, Selective forwarding, Sybil, Sinkhole, Botnet, Hello-flood, MITM	These flaws allow hackers to remotely take control of IoT devices and give them incorrect instructions.
Application Attacks	Code injection, Buffer overflow, SQL injection, Session Hijacking, Authentication and Authorization	As long as hackers can access the application layer, they can steal sensitive data in the IoT.
Zigbee Attacks	Eavesdropping, Replay, Packet forging	In the process of these attacks, attackers will be able to gather

		sensitive data as well as Zigbee traffic.
Z-Wave Attacks	Downgrade, Injection, MITM	Through these attacks, attackers will be able to compromise the security of ZWave devices.

2.6. Intrusion Detection System (IDS)

An IDS is defined as an effective security technology, which can detect and possibly react to computer-related malicious activities (Yaacoub et al., 2023). An ID monitors and analyzes statistics of network activities for potential intrusion and security attacks (Khan & Kim, 2021). IDS can be used to detect different types of malicious network communications and computer systems usage whereas conventional techniques such as firewalls are easily vulnerable to attack and often prone to errors in case of wrong configuration or ambiguous security policies (Haque, 2023).

An IDS can be implemented using two distinct types: signature-based detection and anomaly-based detection. A signature-based IDS uses a database of existing attack signatures and compares the incoming traffic with the database, meaning that an attack can be detected only if the signature is already available in the database. An anomaly-based IDS on the other hand monitors network traffic and attempts to identify any traffic that is abnormal in regards to the normal network traffic. The signature-based detection approach has a major flaw as signature-based IDS will always be susceptible to a zero-day attack or an attacker that modifies the attack to hide from the signature database (Antunes et al., 2022). Anomaly-based IDS are much better suited to use ML as the IDS can be trained to detect the difference between normal traffic and attack traffic (Sharma & Mishra, 2022).

2.7. Machine Learning Algorithms

Machine learning is a field of Artificial Intelligence that is concerned with the design and development of algorithms that allow computers to learn with the help of example data (Chua & Salam, 2023). A major motive in machine learning is to automatically learn to recognize complex patterns and rules so that it can make intelligent decisions based on the given data and past experiences. While many individuals equate Machine Learning (ML), Deep Learning (DL), and Artificial Intelligence (AI), these concepts each possess distinct characteristics, despite their apparent similarities (Isuwa et al., 2022b). AI involves the emulation or replication of human intuition, enabling machines to mimic human thought and behavior. ML, on the other hand, consists in giving an algorithm or in this case a model of a dataset which will be used to identify patterns that can be used to make predictions with future data. DL is a modern technique that uses layers of neurons to simulate the way the human brain processes information and makes decisions. It may be used in applications like predictive analytics and facial recognition.

In the context of intrusion detection, a detection model learns from previously recorded attack patterns (called signatures) and detects similar ones in incoming traffic that have not been encountered previously (Kasongo & Sun, 2020). In this work, our focus is limited to the use of supervised machine learning methods more specifically, Artificial Neural Network (ANN). The subsequent section includes two ML algorithms, including the proposed ANN, and Random Forest (RF) Algorithm.

2.7.1. Artificial Neural Network (ANN) algorithm

ANN is a computational model inspired by the human brain's structure and function. They consist of interconnected layers of artificial neurons that process input data to produce output

(Xu et al., 2023). Through a process known as training, ANN learns to recognize patterns and relationships in data, making them powerful tools for tasks such as classification (Xu et al., 2023). ANN is composed of layers as shown in Figure 2.8: an input layer, one or more hidden layers, and an output layer. Each layer consists of interconnected artificial neurons or nodes, and the connections between them are weighted (Jiang & Xu, 2022). During training, the network learns by adjusting these weights based on the input data and the desired output, to minimize a predefined loss function, a process known as backpropagation (Jiang & Xu, 2022). As training progresses, the network gradually improves its ability to map input data to output prediction accurately.

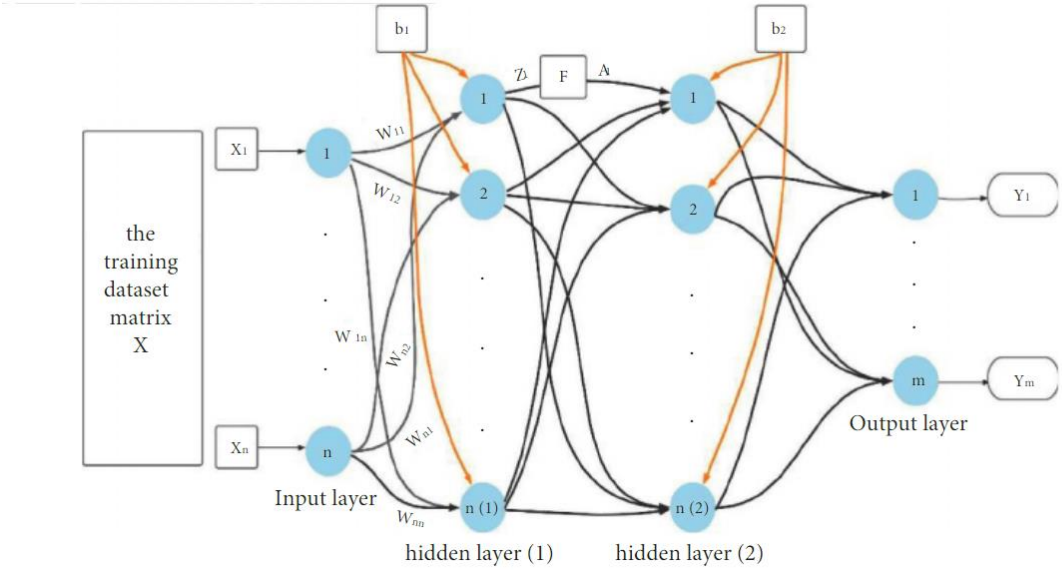

Figure 2.8: Conventional structure of an ANN with two hidden layers (Beghriche et al., 2021)

2.7.2. Random Forest (RF) algorithm

RF is an effective and flexible ensemble learning technique that can be utilized for both regression and classification tasks (Bakar et al., 2022). It constructs multiple decision trees during

the training phase and produces the average prediction for regression or the mode for classification from each tree (Poojitha & Mahaveerakannan, 2023). To minimize the possibility of overfitting and aid in the de-correlation of the trees, RF trains each decision tree with a random subset of the training data and a random subset of the features. The final output of RF is computed during the prediction phase by aggregating the predictions made by each tree. For classification tasks, a majority vote is employed, while for regression tasks, the mean of each tree's predictions is used (Prasanna & Veeramani, 2023).

RF is an effective model known for its ability to handle high-dimensional data, provide high accuracy, and minimize overfitting in large datasets. It also provides estimations of feature importance, aiding in understanding data relationships. However, it can be computationally intensive (Bakar et al., 2022) and may not perform optimally on datasets with substantial feature interactions (Poojitha & Mahaveerakannan, 2023).

2.7.3. Ensemble soft voting classifier

Ensemble-based learning involves using multiple learning algorithms to create a model with superior efficiency and accuracy compared to individual models (Tama & Lim, 2021). The primary goal of ensemble techniques such as Bagging, Stacking, Boosting, and Voting is to achieve better predictive performance by combining the strengths of different classifiers (Taz et al., 2021). Each method has its unique implementation but shares the common objective of improved overall prediction accuracy through a collective approach.

A Voting Classifier, an essential ensemble technique, combines the predictions from multiple base classifiers, which can be either theoretically similar or different, to make a final prediction (Golchha et al., 2022). This method averages the results from each classifier to determine the

class with the highest likelihood of being the output. The concept is to build a single model that learns from various models and predicts output based on their aggregate majority of votes for each output class, rather than building separate specialized models and determining the accuracy for each of them. Two main voting classifiers types are Hard Voting and Soft Voting (Mijoya et al., 2023).

Hard Voting, or majority voting, predicts the output class based on the mode of the predicted class labels from each classifier. The predicted output class is a class with the highest majority of votes, i.e., the class with the highest probability of being predicted by each classifier (Mijoya et al., 2023). This is illustrated in Figure 2.3. The class label y is predicted using the formula in Equation 2.3.

$$\hat{\gamma} = mode\,\{C1(x), C2(x), C3(x), \ldots, C_n(x)\} \tag{2.3}$$

Soft Voting, conversely, involves averaging the predicted probabilities for each class from all classifiers and selecting the class with the highest average probability (Tama & Lim, 2021). Figure 2.2 depicts an illustration of a soft voting classifier. The prediction for the class label $\hat{\gamma}$ is made by averaging the probabilities from each classifier. The class label y is predicted using the formula in Equation 2.4.

$$\hat{\gamma} = \frac{1}{n} \sum_{i=1}^{n} P_i(x) \tag{2.4}$$

Soft Voting generally produces better results than Hard Voting as it accounts for the confidence levels of the classifiers, thus providing more detailed information about the data (García-Ordás et al., 2021). This method offers greater insight into the uncertainty of each prediction and tends to be more robust to noise and outliers by incorporating information from multiple classifiers

to make a more informed decision (Kumari et al., 2021). Figures 2.9 illustrate the working of a hard and soft voting classifier.

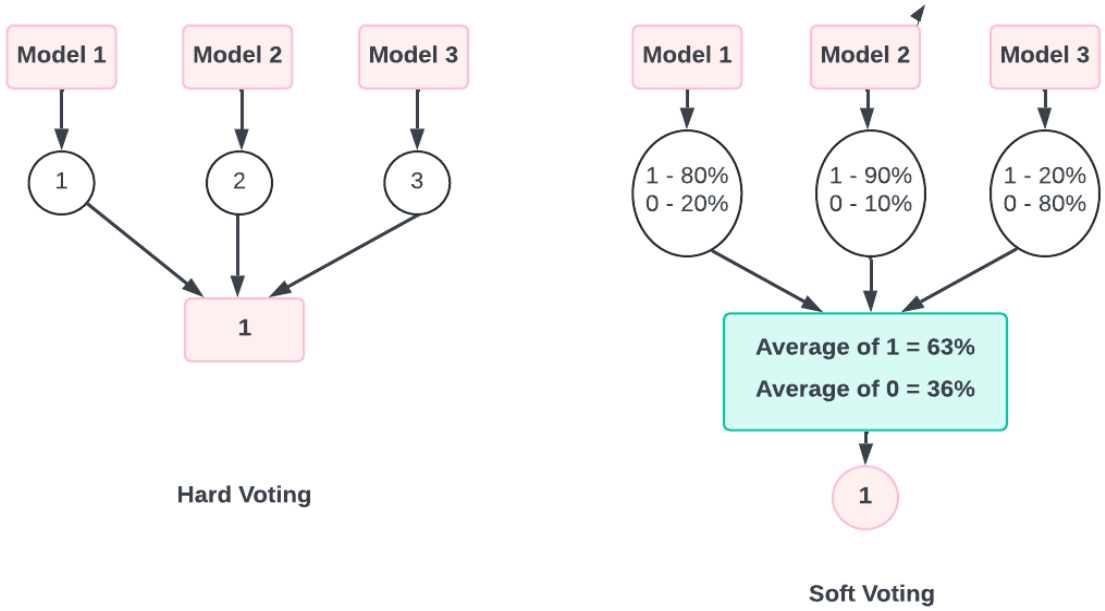

Figure 2.9: An illustration of a hard and soft classifier

2.8. Related Works

Several research works use ML IDS in IoT. These works aim to develop better algorithms and technologies to detect malicious activity on networks, systems, and applications connected to the Internet of Things. Research works have developed a range of techniques such as using supervised and unsupervised learning techniques, deep learning techniques, and statistical methods.

In their study, (Kumar & Singh, 2016) explored the development of an enhanced Intrusion Detection System (IDS) by focusing on two critical aspects: feature selection and the design of an efficient learning algorithm. They introduce an updated version of the Naive Bayes classifier, aiming to outperform existing classifiers like Naive Bayes, J48, and REPTree in both accuracy

and processing time. To evaluate the performance of their proposed classifier, they applied three standard feature selection methods and conduct experiments using the NSL-KDD dataset. Through 10-fold cross-validation, the empirical results demonstrate that the enhanced Naive Bayes classifier achieves superior results in terms of intrusion detection accuracy and a lower false alarm rate compared to the existing methods.

Imad et al., (2022) compared the existing works in terms of the dataset used, the attack/intrusion category, and the obtained results. Subsequently, a proposed model was implemented, wherein the significant features from the Network Intrusion Detection dataset were extracted using Random Forest (RF). The purpose of this extraction was to identify and categorize the data instances into anomalous and normal based on the extracted features. They applied five Machine Learning approaches (NB, DT, KNN, ANN, and LR). The outcomes demonstrated how well the DT model performed, achieving the best accuracy of 100% among all the classifiers. The authors got a perfect model in binary classification that can detect the existence of the attack without any specification about the type of this attack. Additionally, they used one dataset to evaluate their experiment.

A. Das et al., (2022) mentioned the pros and cons, the used dataset, classification method, feature selection method, and preprocessing of each work in their comparative analysis. Besides this, they build their IDS model, where Ensemble Feature Selection, Ensemble Machine Learning classifiers, and three datasets are used to achieve higher accuracy with a lower False Positive Rate. They used majority voting to select the best features subset (EnFs) using ANOVA, Chi2, Lasso, LRL1, MutInfo, Pearson, RF, RFE, and SFPR techniques. They then applied LR, DT, NB, NN, and SVM classifiers to detect intrusions in NSL-KDD, UNSW-NB15, and CICIDS2017 datasets. The absence of the multiclass classification was notable in this study.

In their performance analysis of Machine Learning algorithms in IDS, Saranya et al., (2020) presented a survey of intrusion detection using ML algorithms. Next, a comparative analysis was conducted to evaluate the performance of various classifiers, including Linear Discriminant Analysis (LDA), Classification and Regression Trees (CART), RF, SVM, NB, DT, and Logistic Regression (LR). The experimentation was performed using the KDDcup99 dataset. The findings revealed that the RF classifier achieved the highest accuracy level, reaching 99.81%. This notable performance can be due to the utilization of a filter method that effectively reduced the number of features from 42 to 20. However, it is important to note that the proposed method was not specifically investigated for multiclass classification in this study.

Pajouh et al., (2020) they presented an intrusion detection model based on a two-layer dimension reduction and two-level classification module. This model was also designed to identify malicious activities such as User to Root (U2R) and Remote Local (R2L) attacks. For dimension reduction, component analysis and linear discriminant analysis were used. The NSL-KDD dataset was used to perform the entire experiment. Naive Bayes and the Certainty Factor version of K-Nearest Neighbor were applied to detect suspicious behavior with the two-level classification module.

Mukherjee & Sharma, (2012) showed the importance of feature selection in building effective and efficient intrusion detection systems. They proposed a feature vitality-based data reduction approach (FVBRM) to identify a reduced set of important input features using the NSL-KDD dataset.

CHAPTER THREE

METHODOLOGY

3.1. Introduction

This chapter describes the research method utilized in the study, the architecture design, the parameters, and settings used, and the benchmark dataset employed. It concludes with a description of the experimental design for presenting results.

3.2. Dataset Description

In this study, the NSL-KDD dataset will be used for the evaluation of the proposed model. This dataset has been utilized as a benchmark dataset by numerous researchers in the field of IDS (Chakrawarti & Shrivastava, 2023; Sahli, 2022; Sarthak Rastogi et al., 2022; Thakkar & Lohiya, 2021). The NSL-KDD dataset is made up of 41 attributes, along with a class label attribute classifying each record, and 125,973 instances. It is divided into a training set (80%) and a test set (20%). The NSL-KDD dataset divides the network traffic as normal and anomalous. It consists of five classes, one of which is normal, and the other four are types of attacks. Table 3.1 shows the number of instances and percentage of normal and attack cases in each training and testing dataset fold.

Table 3.1: Number of attack cases in the training and testing dataset (Oche et al., 2021).

Dataset	**Number of Instances**					
	Normal	**DOS**	**Probe**	**U2R**	**R2L**	**Total**

Training set	67,343 (53%)	45,927 (37%)	11,656 (9.11%)	52 (0.04%)	995 (0.85%)	**125,973**
Testing set	9,711 (43%)	7,456 (33%)	2,421 (11%)	200 (0.9%)	2,656 (12.1%)	**22,544**

Within the data set exists 4 different classes of attacks: Denial of Service (DoS), Probe, User to Root (U2R), and Remote to Local (R2L). Table 3.2 shows the attack types of each attack class.

Table 3.2: Attack types in the NSL KDD Dataset

Attacks in Dataset	**Attack Type (37)**
DoS	Back, Land, Neptune, Pod, Smurf, Teardrop, Mailbomb, Processtable Udpstorm, Apache2, Worm
Probe	Satan, IPsweep, Nmap, Portsweep, Mscan, Saint
R2L	Guess_password, Ftp_write, Imap, Phf, Multihop, Warezmaster, Xlock, Xsnoop, Snmpguess, Snmpgetattack, Httptunnel, Sendmail, Named
U2R	Buffer_ overflow, Loadmodule, Rootkit, Perl, Sqlattack, Xterm, Ps

3.3. Proposed Method Design

Figure 3.1 presents the overall architecture of the proposed method. Afterward, a step-wise description of the stages and processes involved is provided.

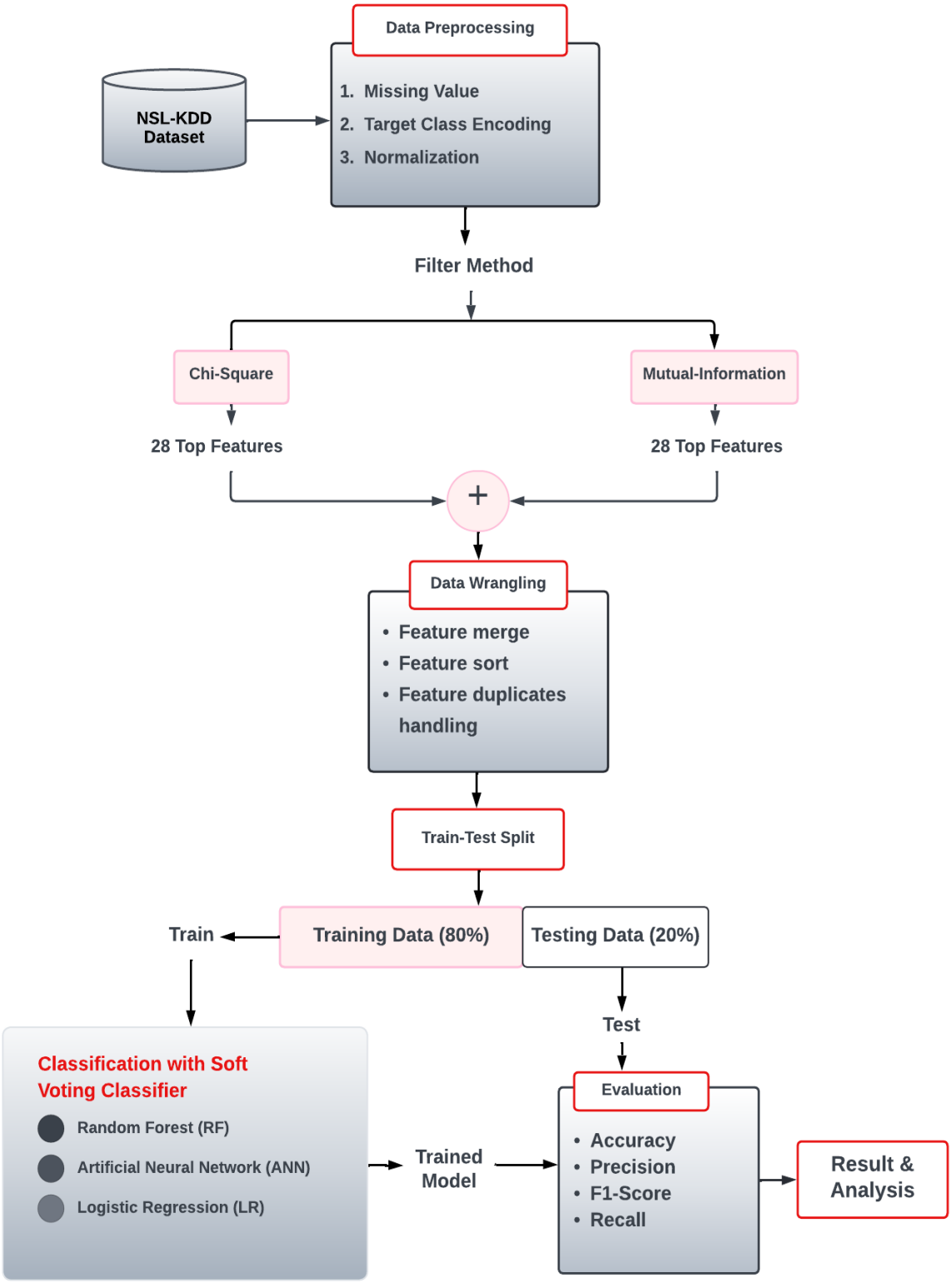

Figure 3.1: The Proposed Method Architecture

3.3.1. Data Preprocessing

To prepare the NSL-KDD dataset for model training, adequate preprocessing is essential. In this study, data preprocessing was conducted independently on the training and testing datasets, as they were provided in separate CSV files. This approach ensured that preprocessing steps were tailored to the distinct characteristics of each dataset, thereby preserving the integrity and independence of the training and testing phases. Handling missing values is the first step in preprocessing, ensuring the dataset is complete and reliable. This involves checking for any missing data points and addressing them. Upon examination, it was discovered that the dataset had no missing values.

The multi-class target variable is then transformed into a binary format during the target class encoding process by merging the classes into two categories: normal and abnormal. This transformation is necessary because it simplifies the classification task, making it easier for the ML model to differentiate between normal and anomalous network activity. The target class is encoded with the labels "0" for normal network activities and "1" for abnormal activities. Figures 3.2 and 3.3 depict a bar chart showing a visualization of the distribution of labels in the training and testing sets respectively.

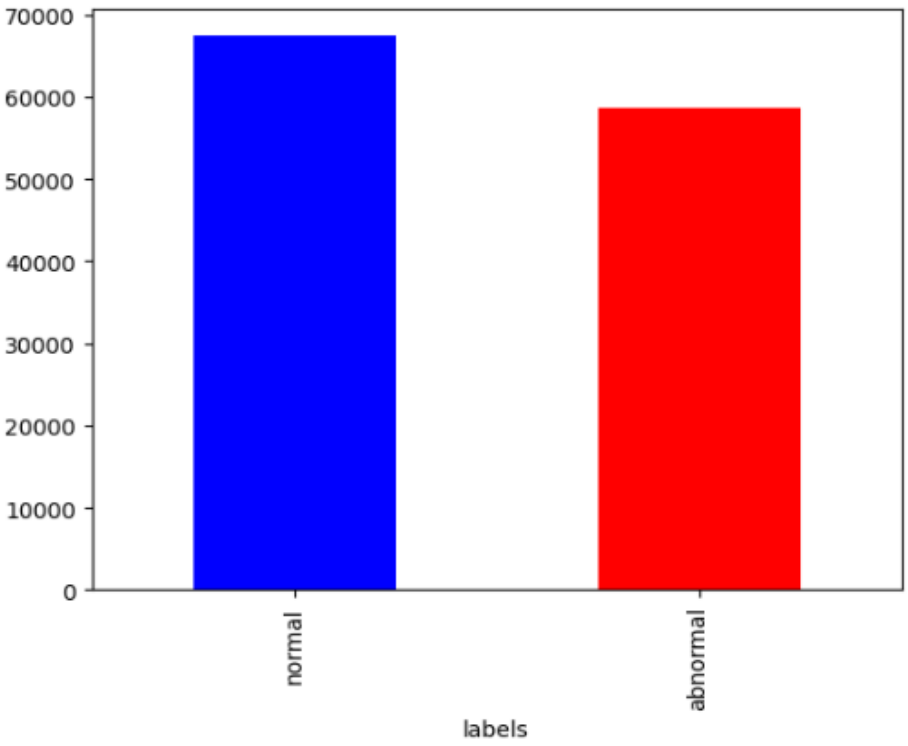

Figure 3.2: Visualization of the distribution of labels in the training set

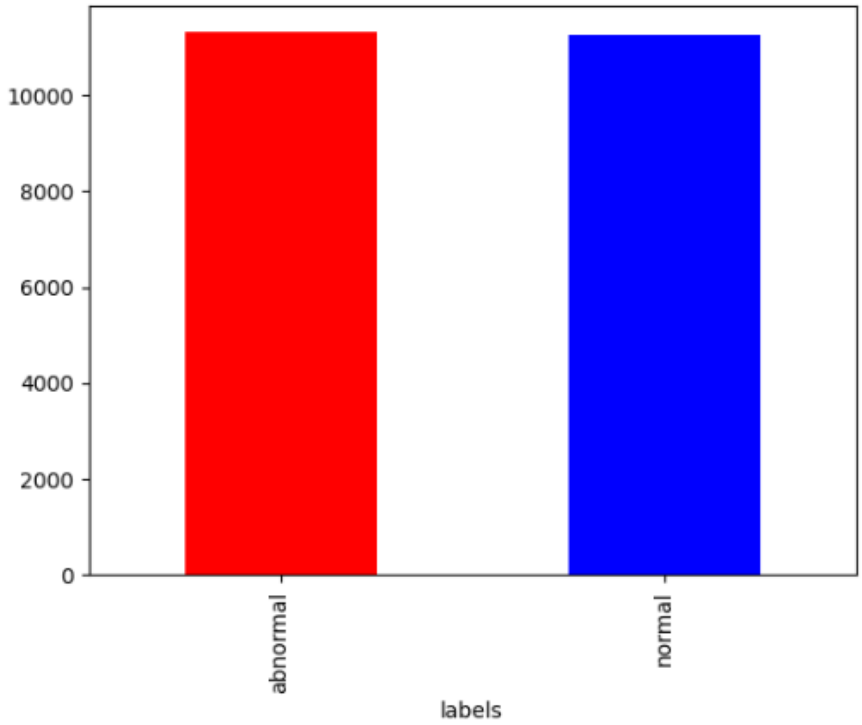

Figure 3.3: Visualization of the distribution of Labels in the testing set

Next, the features are normalized to ensure input variables are scaled to a comparable range. Normalization is a crucial process that adjusts feature values to a common scale while maintaining differences within value ranges. Normalizing the input variables ensures that no single feature dominates the learning process due to its scale, leading to faster training and improved model performance. This is particularly important for algorithms like neural networks that are sensitive to data size. By bringing all features to a common scale, normalization helps the model learn patterns more effectively and makes the training process more efficient and reliable.

3.3.2. Feature Selection

Feature selection focuses on identifying the most relevant attributes that contribute to detecting network intrusions. In this study, two filter methods are used: chi-square and mutual information. These methods were chosen for their effectiveness in handling large datasets and their ability to rank attributes based on their importance to the target class. The chi-square method assesses the independence between each feature and the target variable, while mutual information measures the amount of information each feature provides about the target variable. Afterward, the top 70% of the attributes, amounting to 28 features, are selected from each of the two methods. This helps to identify features that are most relevant and ensures that the model is more equipped to identify patterns and make accurate predictions. Figures 3.4 and 3.5 depict the features selected by chi-square and mutual information respectively.

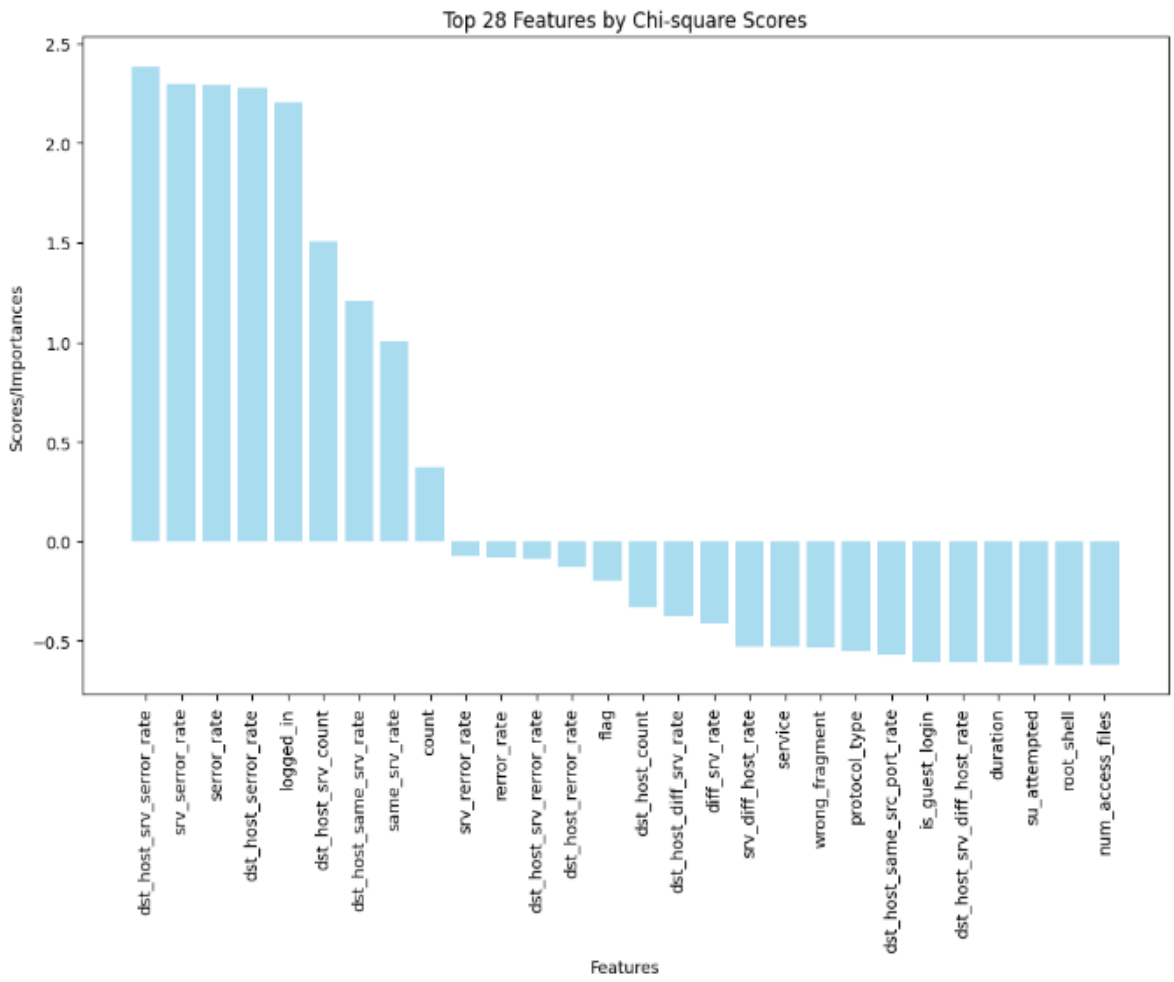

Figure 3.4: Top 28 Selected Features by Chi-Square

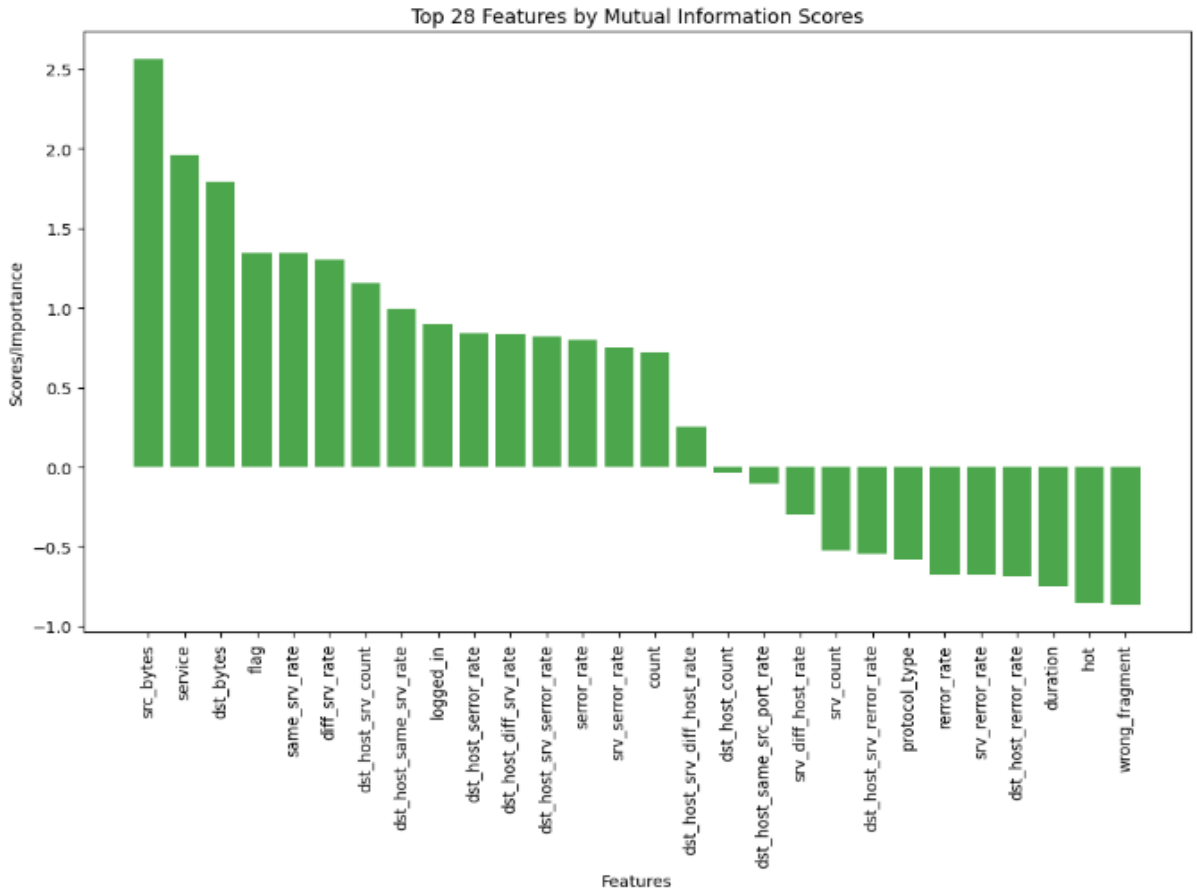

Figure 3.5: Top 28 Selected Features by Mutual Information

3.3.3. Data Wrangling

A single set of relevant features is created by combining the top 28 features from the two filter methods. This is carried out to further refine the feature selection process. After the merging process, the features are ranked based on their scores, with the most relevant feature carrying the highest score. Any duplicates, i.e., features chosen by both methods, are then eliminated. To guarantee a distinct collection of top features, duplicates that come in second place in the ranking are discarded. The end deliverable is a dataset with features that maximally contribute to the

model prediction. Figure 3.6 depicts a graph of the selected features from both filter methods and their respective rankings.

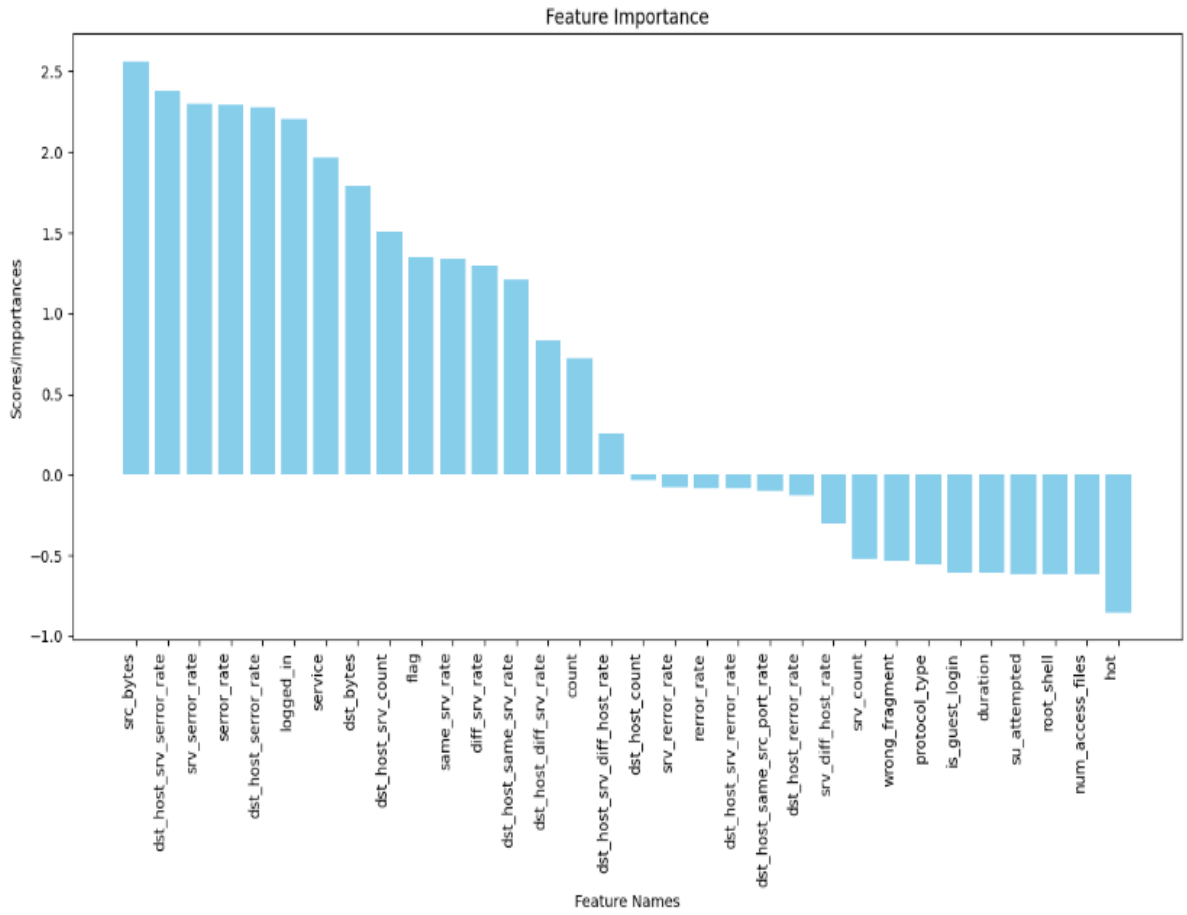

Figure 3.6: Selected Features by Chi-Square and Mutual Information after merging

3.3.4. Train-Test Split

To maintain consistent data distribution and representation, the train-test split combines the training and testing datasets from the original datasets into a single dataset. The merged set is then split again into distinct training and testing sets, which are then fed to the learning algorithm. This approach minimizes possible biases and ensures that both sets have similar statistical

characteristics, which is essential for an unbiased evaluation of the model and improves its ability to generalize and be more reliable in practical situations.

3.3.5. Classification with Soft Voting Classifier

In this study, three distinct classifiers; ANN, RF, and LR are utilized and combined into a soft voting classifier. Each classifier is chosen for its unique strengths.

ANN is a powerful model capable of capturing complex patterns and relationships within data. They consist of multiple layers of interconnected neurons that process input features and learn from the data. ANNs are particularly effective in recognizing intricate patterns that may not be apparent through linear relationships, making them suitable for detecting sophisticated network intrusions.

RF is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of their predictions. This approach enhances the model's stability and accuracy by reducing overfitting and improving generalization. RF is robust to noise and can handle large datasets with high dimensionality, making it an excellent choice for intrusion detection.

LR is a simple yet effective linear model used for binary classification tasks. It estimates the probability of a binary outcome based on the input features, providing a clear baseline for comparison with more complex models. LR is valued for its interpretability and efficiency in binary classification tasks, such as distinguishing between normal and abnormal network activities.

The predictions from these classifiers are combined using a soft voting classifier. In soft voting, the output from each of these algorithms is combined through a weighted average, where the

weights are determined during the training phase based on the individual performance of each algorithm on the dataset. This ensemble approach leverages the strengths of each classifier, improving overall detection accuracy and reliability. By combining the predictive power of ANN, RF, and LR, a higher level of performance is achieved.

3.3.6. Model Performance Evaluation

Evaluating a model is an essential part of building an effective machine-learning model. The confusion matrix depicted in Table 3.3 is the basis on which the performance of the model and calculation of the various evaluation metrics is built. The following metrics were calculated to evaluate the performance of the developed system.

Table 3.3: Confusion matrix

Class		Predicted Class	
		Normal	Abnormal (attack)
Actual class	Normal	TP	FN
	Abnormal (attack)	FP	TN

Where TP, TN, FP, and FN stand for true positives, true negatives, false positives, and false negatives respectively. The following are the metrics used for the evaluation of the proposed model.

i. **Accuracy:** Accuracy measures the overall correctness of predictions by calculating the ratio of correctly predicted instances (both true positives and true negatives) to the total number of instances. It can be calculated as depicted in Equation (3.1)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.1}$$

ii. **Precision:** Precision quantifies the exactness by measuring the proportion of true positive predictions among all positive predictions. It can be calculated as depicted in Equation (3.2)

$$Precision = \frac{TP}{TP + FP} \tag{3.2}$$

iii. Recall: Recall, also known as sensitivity or true positive rate, measures the ability to identify actual positive instances among all actual positive instances. It can be calculated as depicted in Equation (3.3)

$$Recall = \frac{TP}{TP + FN} \tag{3.3}$$

iv. F1 Score: The F1 score is the harmonic mean of precision and recall, providing a balanced assessment of the model's performance. It is a weighted average of the precision and recall of a model. It can be calculated as depicted in Equation (3.4)

$$F1 = 2 \times \frac{precision \times recall}{precision + recall} \quad = \frac{2 * TP}{2 * TP + FN + FP} \tag{3.4}$$

3.4. Parameter Values and Settings

Table 3.4 lists the parameters and settings used in the experiments for each of the combined learning algorithms, along with additional pertinent configurations.

Table 3.4: Parameter values and settings utilized in the proposed method

Random Forest Classifier	$\text{n_estimators} = 150$
Logistic Regression	Max_iter = 1000
Artificial Neural Network	$hidden_layer_sizes = (32, 8)$, $\text{early_stopping} = \text{False}$, $\text{solver} = 'adam'$, $\text{nesterovs_momentum} = \text{True}$, $\text{momentum} = 0.90$, $\text{max_iter} = 500$
Other Settings	$train_test_spilt = 80{:}20, random_state = 42,$

3.5. Materials

The experiments in this study are conducted in a Jupyter Notebook Integrated Development Environment (IDE). The implementation process is performed using the Python programming language on a computer that has an Intel (R) Core (TM) i5-5300U CPU with a processing speed of 2.40 GHz and a RAM capacity of 4.00 GB.

3.6. Experimental design

The experiments and result presentation in this study involve a thorough evaluation of the model using both the complete set of features and a refined subset of features from the NSL-KDD dataset. The experiment will be conducted in Three distinct forms;

i. The model will be trained using all available features in the dataset to establish a baseline performance, which is then assessed using key metrics such as accuracy, precision, recall,

and F1 score. This phase helps in understanding the model's effectiveness without any feature reduction.

ii. The model then undergoes training with a selected subset of features obtained through chi-square and mutual information methods, aiming to enhance performance by focusing on the most relevant attributes. The performance is re-evaluated using the same metrics to facilitate a direct comparison with the baseline results.

iii. Finally, the study compares these findings with existing works in the field to validate the proposed approach and highlight any advancements or improvements.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1. Introduction

This chapter presents the results of experiments and provides a detailed discussion of the performance of the proposed Intrusion Detection System (IDS) using the feature selected and the entire dataset. The chapter is segmented into three main sections: the first section discusses the results obtained using the feature-selected dataset, the second section presents the results using the entire dataset without feature selection, and the third section compares the results from both experiments and contrasts them with existing works of literature.

4.2. Experiment with Feature Selection

The first phase of the experiment involves using the dataset with selected features. The results demonstrate the effectiveness of feature selection in improving the model's performance. The accuracy of the model was found to be 99.40%, indicating that the model correctly identified the majority of the instances. Precision was recorded at 99.12%, which signifies that the model has a high rate of correctly identifying true positive instances among the detected positives. Recall, a measure of the model's ability to identify all relevant instances, stood at 99.76%, reflecting its efficiency in detecting almost all intrusion attempts. The F1 score, which balances precision and recall, was 99.44%, highlighting the model's overall robustness and reliability in detecting intrusions. These results suggest that the feature selection method significantly enhances the detection performance of the IDS by focusing on the most relevant attributes. Table 4.1 below summarizes these performance metrics:

Table 4.1: Performance Results for the Experiment with Feature-Selection

Metric	**Value (%)**
Accuracy	99.40
Precision	99.12
Recall	99.76
F1 Score	99.44

In addition to the performance metrics, the confusion matrix depicted in Figure 4.1 was utilized to provide a deeper understanding of the model's classification capabilities. The confusion matrix shows that the model accurately classified a significant number of intrusions as intrusions (52.92%) and normal traffic as normal (46.55%), with very few misclassifications.

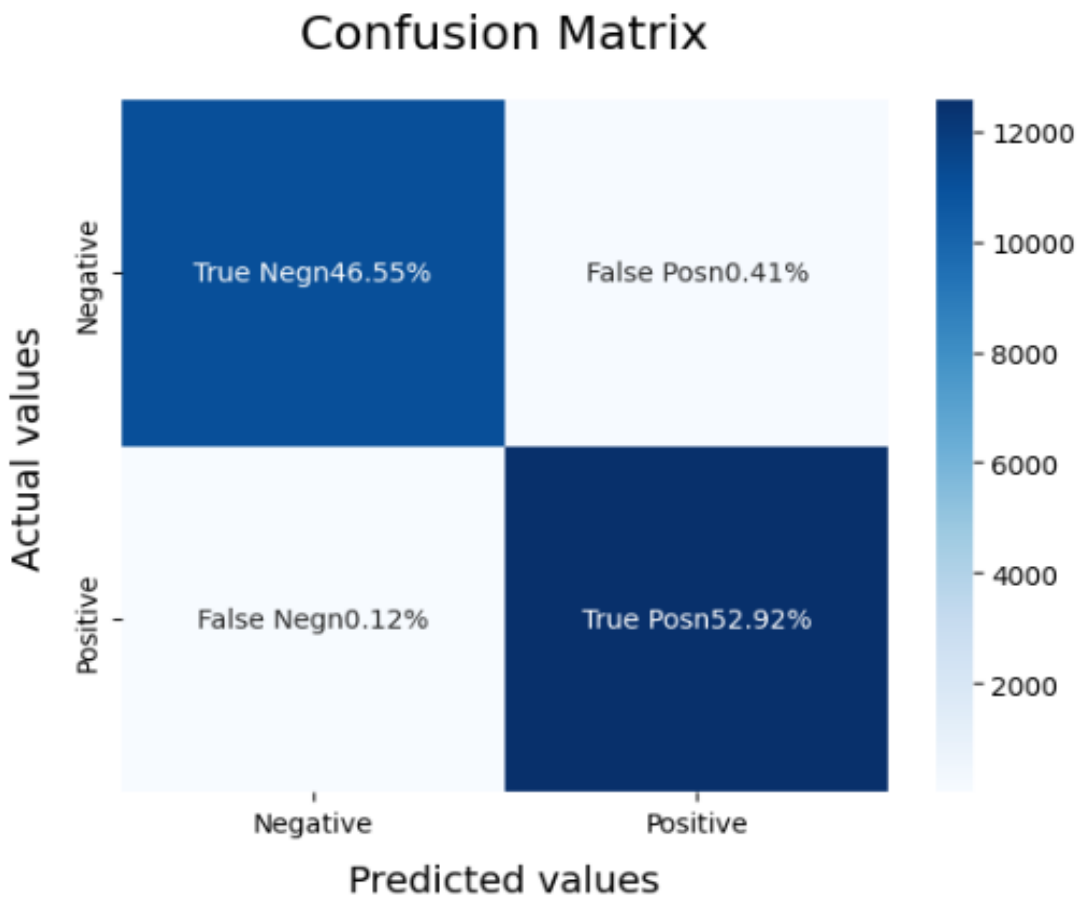

Figure 4.1: Confusion Matrix for the experiment with Feature Selection

4.3. Experiment with Entire Dataset

The second phase of the experiment involved training and testing the model using the entire dataset without any feature selection. This approach aimed to evaluate the model's performance with all available attributes and compare it with the feature-selected approach.

The results from this experiment were slightly lower than those obtained using the feature-selected dataset. The model achieved an accuracy of 99.28%, a precision of 99.01%, a recall of

99.63%, and an F1 score of 99.32%. Although these metrics are still high, they indicate a marginal decline in performance compared to the feature-selected dataset.

The slightly lower performance metrics suggest that using all features, including irrelevant or redundant ones, may introduce noise and reduce the model's efficiency. This emphasizes the importance of feature selection in enhancing the performance of machine learning models by eliminating less informative attributes and focusing on the most critical ones. Table 4.2 below provides a summary of the performance metrics for the entire dataset experiment:

Table 4.2: Performance Results for the entire Dataset Experiment

Metric	**Value (%)**
Accuracy	99.28
Precision	99.01
Recall	99.63
F1 Score	99.32

4.4. Comparison and Discussion

The final section of this chapter compares the results obtained from the feature-selected dataset experiment and the entire dataset experiment, as well as between the proposed method feature-selection experiment and existing studies. This comparison aims to highlight the impact of feature selection on the model's performance and validate the proposed methodology's effectiveness.

When comparing the two results from the phases, it is evident that the feature-selected dataset experiment outperformed the entire dataset experiment in all evaluation metrics. The feature selection process enhanced the model's accuracy, precision, recall, and F1 score, demonstrating the importance of feature selection in enhancing the model's efficiency by reducing dimensionality and focusing on the most relevant features. Table 4.3 below summarizes the comparison of the performance between the feature-selected and entire dataset experiments:

Table 4.3: Performance Results Between the Feature selected and Entire Dataset

Dataset	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 Score (%)**
Feature-Selected Dataset	**99.40**	**99.12**	**99.76**	**99.44**
Entire Dataset	99.28	99.01	99.63	99.32

To further illustrate this comparison, a bar chart showing the performance of both experiments is given in Figure 4.2. This visualization helps to clearly show the superior performance of the feature-selected approach.

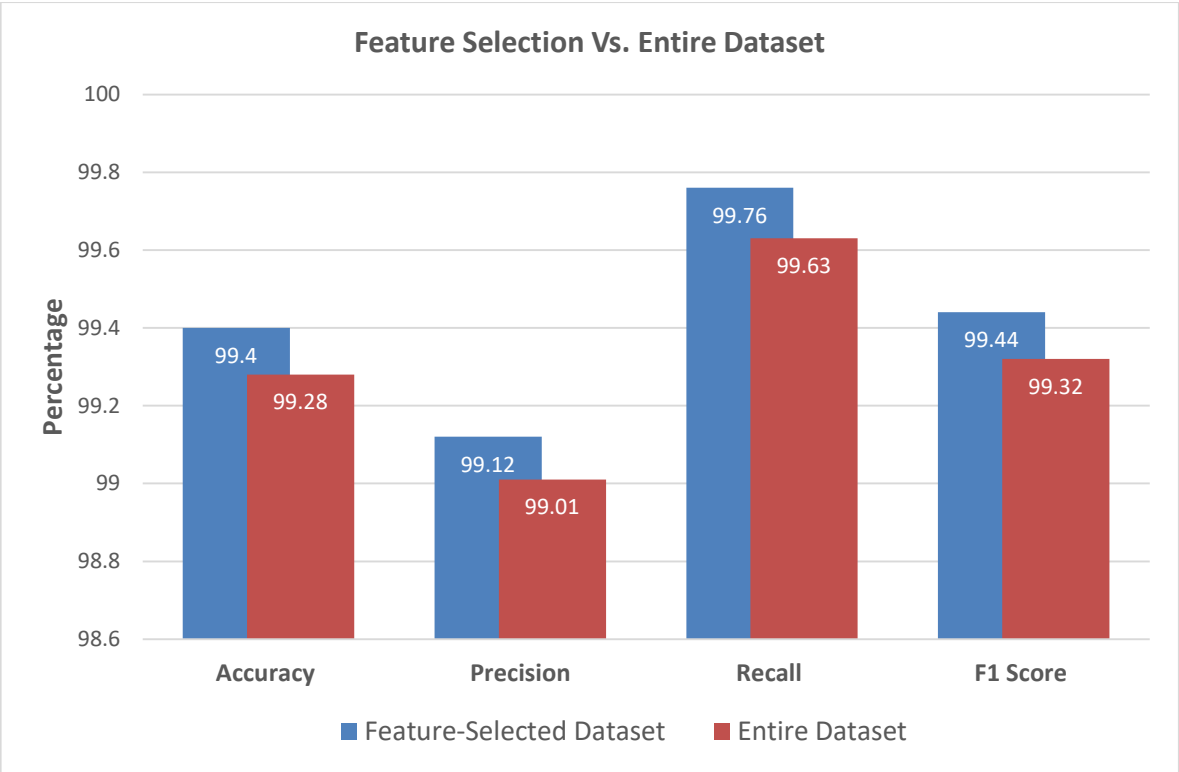

Figure 4.2: Bar Chart Comparison Between Feature Selected and Entire Dataset

Next, we compare the results from the feature-selected dataset experiment with three recent studies reviewed in the literature. The basis of this comparison is the use of feature selection and the NSL-KDD dataset.

Kumar and Singh (2016) employed three feature selection techniques, with the Gain Ratio using the ranker method yielding the best results: 92.33% accuracy, 85.89% recall, 98.00% precision, and a 99.00% F1 score. When compared to these results, the proposed method demonstrates superior performance across all metrics, particularly in the use of feature selection.

Other existing studies in the literature include; Das et al., (2022), who used Recursive Feature Selection (RFE) and achieved 96.21% accuracy, 96.00% recall, 95.00% precision, and 96.41% F1 score. Pajouh et al., (2020) employed Two-layer Dimension Reduction (TLD) and reported

an accuracy of 85.03% and a recall of 84.86%. Comparing these results to that of my proposed method, it is evident that the use of the Chi-square and Mutual Information filter method in the proposed method gives a better performance surpassing those in the other studies reviewed in the literature, highlighting its robustness and efficiency in detecting intrusions in IoT networks.

The Proposed method outperforms other existing studies, demonstrating the effectiveness of our feature selection and classification techniques.

Table 4.4 summarizes the comparison with these existing works. Note that in the table to be presented, there are instances of missing results/values, particularly in Pajouh et al., (2020) work, meaning that the author did not employ the use of those matrices in their study. Thus the comparison is only carried out with the scores that are recorded. These missing values will be replaced with a dash.

Table 4.4: Comparison of the Proposed Method with Existing Studies in the Literature

Study	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 Score (%)**
Proposed Method	**99.40**	**99.12**	**99.76**	**99.44**
Kumar & Singh, (2016)	92.33	85.89	98.00	99.00
Das et al., (2022)	96.21	96.00	95.00	96.41
Pajouh et al., (2020)	85.03	84.86	—	—

To further illustrate this comparison, a bar chart showing the performance of the proposed method with the other existing works in literature including the baseline paper is depicted in Figure 4.3. This visualization highlights how my proposed method surpasses previous works in terms of accuracy, precision, recall, and F1 score.

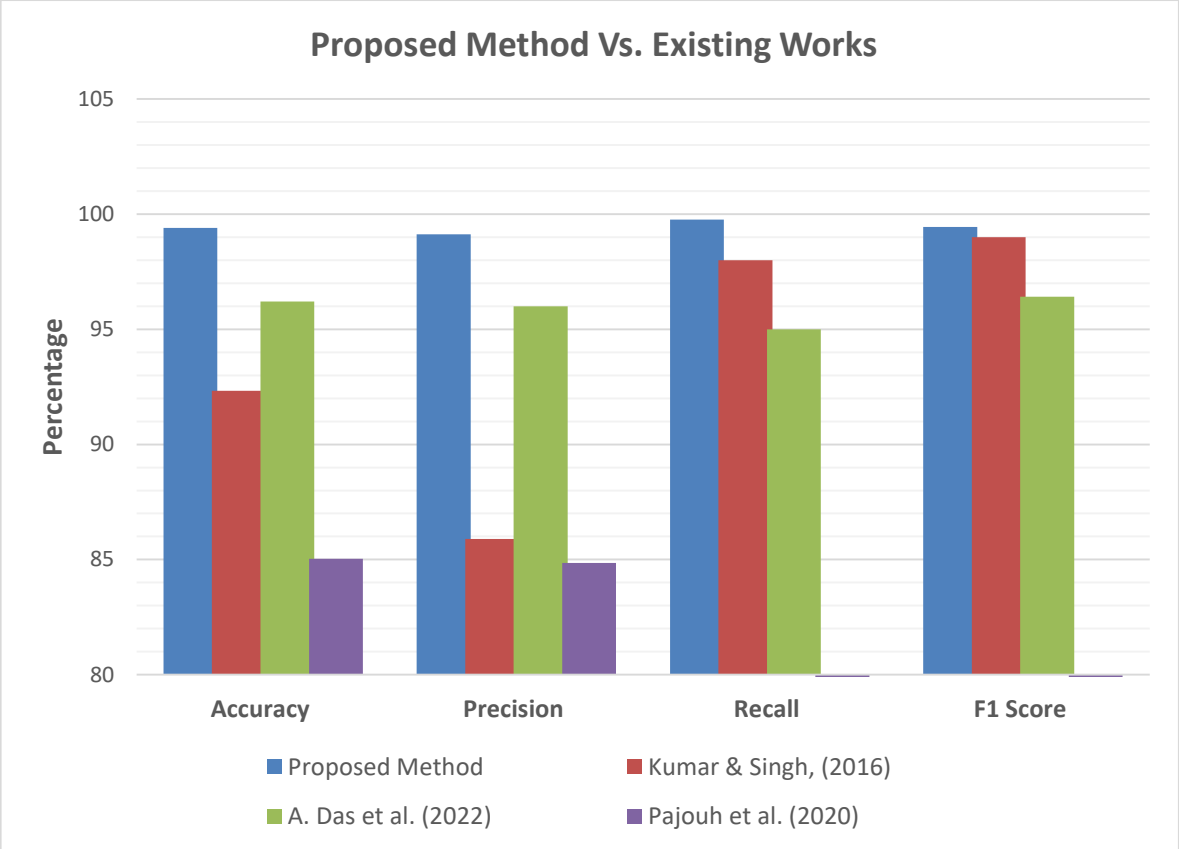

Figure 4.3: Bar Chart Comparison of the Proposed Method with Existing Studies

The proposed method demonstrates a significant improvement over existing studies in accuracy, precision, recall, and F1 score, largely due to a meticulous feature selection process and the use of an ensemble soft voting classifier. By applying Chi-square and Mutual Information filters to the NSL-KDD dataset, the most important features were retained, reducing data dimensionality and allowing the classifier to focus on the most informative characteristics, thus enhancing intrusion detection. The ensemble classifier, which combined the strengths of three distinct algorithms, ensured that the final prediction was a balanced consensus, avoiding over-reliance on any single model. This, along with careful tuning and refinement of model parameters during training, greatly enhanced the model's predictive capability and performance on unseen data.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1. Summary

This research aimed to address the escalating security challenges in Internet of Things (IoT) networks, which have become increasingly vulnerable to sophisticated cyber threats due to their vast, interconnected nature. The inadequacy of traditional Intrusion Detection Systems (IDS) in handling the specific traits in IoT environments necessitated the development of a more effective solution. Consequently, this study focused on creating a machine learning-based IDS tailored specifically for IoT networks, utilizing the NSL-KDD dataset; a widely recognized benchmark in intrusion detection research.

The research methodology was meticulously designed, starting with feature selection using the Chi-square and Mutual Information filter method to identify the most relevant features from the NSL-KDD dataset. A predictive model was then built using a soft voting classifier that combined Artificial Neural Network (ANN), Random Forest (RF), and Logistic Regression (LR), leveraging their complementary strengths to enhance detection performance.

The experimental phase was divided into two sections: one using a dataset with feature selection and the other using the entire dataset, with critical analysis and comparison of results. Additionally, the performance of the feature-selected model was benchmarked against entire dataset experiment and the existing literature. The findings demonstrated the superiority of the proposed model in effectively detecting intrusions within IoT environments.

5.2. Conclusion

This research developed a soft voting classifier-based IDS that provides a robust and highly effective solution to the pressing security issues in IoT networks. The results indicate that the feature selection process was instrumental in improving the IDS's performance, as evidenced by the superior metrics achieved in the proposed method with the selected features; particularly an accuracy of 99.40% and an F1 score of 99.44%. These metrics surpass those achieved when using the entire dataset. The comparative analysis further demonstrates the superior performance of the proposed method over other existing studies. The integration of ANN, RF, and LR within the soft voting framework allowed for a balanced and comprehensive detection capability, making the IDS adaptable to the dynamic and diverse nature of IoT threats.

This study not only contributes to the body of knowledge in IoT security but also provides a practical framework for the development of more advanced and efficient IDS. The successful application of feature selection and the use of a soft voting classifier have demonstrated that it is possible to enhance the security of IoT networks significantly, ensuring both the safety and reliability of these critical systems.

5.3. Recommendations

Based on the findings of this research, several recommendations are proposed for future work and practical implementation. First, it is recommended that the developed IDS be deployed and tested in real-world IoT environments. This would validate the model's effectiveness under actual operating conditions and ensure its adaptability to the ever-evolving landscape of cyber threats. Second, the soft voting classifier could be improved by incorporating additional machine

learning algorithms, such as Support Vector Machines (SVM) or Gradient Boosting Machines (GBM), to enhance accuracy and detection speed.

Additionally, given the rapid evolution of cyber threats, continuous research must be conducted to keep the IDS updated and responsive to new challenges. This could involve periodic retraining of the model with new datasets or the incorporation of adaptive learning techniques that allow the IDS to evolve alongside emerging threats. Collaboration with industry stakeholders could also play a crucial role in refining the system and ensuring its implementation across various IoT sectors, thereby enhancing the security infrastructure of these critical networks.

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